

# MINDFUL EATING: A PATH TO A HEALTHY BODY

## Topic 1: Diversity of Food Across Regions

### Definition

Food diversity refers to the variety of traditional food items consumed in different regions, which depends largely on locally grown crops, climate, and culture.

### Concept Explanation

India is a vast country with different types of soil and climatic conditions. Because of this, different crops grow better in different areas. For example, wheat and maize grow well in Punjab, so their traditional food involves *makki di roti*. In southern states like Karnataka, rice and coconut are abundant, making *idli*, *dosa*, and coconut chutney staple foods. The food choices of a region are deeply connected to agriculture, taste preferences, and traditional practices.

### Diagram



### Real-Life Example

If you travel from the cold mountains of Himachal Pradesh to the coastal areas of Kerala, the everyday meals completely change from wheat-based and warm foods to rice-based and seafood-rich diets.

### ☒ Quick Revision Checklist:

- ☐ Food choices vary from state to state in India.
- ☐ Traditional food is mostly based on locally grown crops.
- ☐ Climate and soil type determine which crops grow in a region.

## Topic 2: Changes in Culinary (Cooking) Practices

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### Definition

Culinary practices are the methods, tools, and traditions used in the preparation and cooking of food.

### Concept Explanation

How we cook our food has changed significantly over time due to technological advancements. In the past, people primarily cooked food on a *chulha* (a traditional stove burning wood or coal) and manually ground spices and pastes using a *sil-batta* (stone grinder). Today, most kitchens use modern gas stoves or induction cooktops, and electrical mixer-grinders do the hard work of grinding in seconds. These changes make cooking faster and more convenient.

### Diagram



### Real-Life Example

Your grandmother might recall spending hours grinding soaked rice and dal on a stone grinder to make *dosa* batter, while today, we can achieve the same result in minutes using an electric blender.

### ☑ Quick Revision Checklist:

- ☐ Culinary practices refer to cooking methods and tools.
- ☐ Traditional tools: *Chulha* (stove), *Sil-batta* (stone grinder).
- ☐ Modern tools: Gas stove, electrical grinder.
- ☐ Technology and better transportation have driven these changes.

### Topic 3: Components of Food (Nutrients)

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## Definition

**Nutrients** are essential components of food that provide energy, support growth, repair the body, and protect against diseases.

## Concept Explanation

The food we eat contains five major nutrients, plus two extra essential components:

1. **Carbohydrates (Energy-giving):** Provide instant energy. Sources: Wheat, rice, potato, sugar, banana.
2. **Fats (Energy-giving):** Stored energy that keeps us warm. Sources: Ghee, oil, nuts, seeds, butter.
3. **Proteins (Body-building):** Help in muscle growth and body repair. Sources: Pulses (dal), soya bean, milk, paneer, eggs, fish.
4. **Vitamins & Minerals (Protective nutrients):** Required in small amounts to protect the body from diseases.
  - *Vitamin A:* For eyes and skin (Papaya, carrot).
  - *Vitamin C:* Fights diseases, prevents scurvy (Lemon, orange, amla).
  - *Calcium:* For strong bones and teeth (Milk, curd).
  - *Iodine:* Prevents goitre (Iodised salt, seafood).
  - *Iron:* Important for blood, prevents anaemia (Spinach, beetroot).
5. **Dietary Fibres (Roughage):** Do not provide nutrients but help get rid of undigested food and prevent constipation. Sources: Fresh fruits, whole grains.
6. **Water:** Helps absorb nutrients and removes waste via sweat and urine.

## Diagram



## Real-Life Example

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Marathon runners drink glucose water during a race because glucose is a carbohydrate that gives *instant* energy. Polar bears accumulate a thick layer of fat under their skin to survive their long winter sleep without eating.

### ☑ Quick Revision Checklist:

- ☐ Carbs & Fats = Energy.
- ☐ Proteins = Growth and repair.
- ☐ Vitamins & Minerals = Protection from diseases.
- ☐ Roughage (Fibre) = Smooth digestion.
- ☐ Water = Nutrient absorption and waste removal.

### Topic 4: Testing for Nutrients in Food

#### Definition

Simple chemical tests can be performed to detect the presence of specific nutrients like starch, fats, and proteins in our food.

#### Concept Explanation

Scientists use specific reactions to find out what is inside our food:

1. **Test for Starch (Carbohydrate):** Add 2-3 drops of dilute iodine solution to the food. If the colour changes to **blue-black**, starch is present.
2. **Test for Fat:** Wrap the food in a piece of paper and gently press it. Let it dry. If a translucent **oily patch** remains on the paper, the food contains fat.
3. **Test for Protein:** Make a paste of the food in a test tube. Add 2 drops of **Copper Sulfate** solution and 10 drops of **Caustic Soda** solution. Shake well. If the mixture turns **violet**, protein is present.

#### Diagram





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### Real-Life Example

If you accidentally drop iodine on a piece of white bread, it will turn blue-black immediately because bread is rich in starch. If you keep a hot *parantha* wrapped in newspaper, the paper becomes slightly transparent due to the ghee/oil (fat test).

#### ☑ Quick Revision Checklist:

- [ ] Starch + Iodine  $\rightarrow$  Blue-Black colour.
- [ ] Fat + Paper  $\rightarrow$  Oily, translucent patch.
- [ ] Protein + Copper Sulfate + Caustic Soda  $\rightarrow$  Violet colour.

### Topic 5: Balanced Diet vs. Junk Food

#### Definition

A **Balanced Diet** contains all essential nutrients, roughage, and water in the correct amounts needed for proper growth. **Junk Food** is food high in calories (from sugar and fat) but severely lacking in essential nutrients, vitamins, and fibre.

#### Concept Explanation

Our body's nutritional requirements change based on age, gender, and physical activity. A growing teenager or an athlete needs a diet rich in proteins and carbohydrates, while an elderly person might need more dietary fibre. A balanced diet perfectly matches the body's needs. On the other hand, frequently eating junk food (like potato chips, candies, and cold drinks) leads to obesity and severe health issues because it provides empty calories without nourishing the body.

#### Diagram



### Real-Life Example

Comparing roasted *chana* to packaged potato wafers: Roasted *chana* has high protein, high fibre, and low fat. Potato wafers have extremely high fat and very low protein/fibre. Eating the *chana* nourishes the body, while the wafers just add unhealthy fat.

#### ☑ Quick Revision Checklist:

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- ☐ A balanced diet provides exact, necessary proportions of all nutrients.
- ☐ Nutritional needs vary by age and physical activity.
- ☐ Junk food causes obesity and lacks vitamins/minerals.

### Topic 6: Millets – The Nutri-Cereals

#### Definition

Millets are small-seeded, highly nutritious native Indian crops (like *jowar*, *bajra*, *ragi*) that can grow in diverse and tough climatic conditions.

#### Concept Explanation

Millets have been a part of the Indian diet for centuries and are making a massive comeback! They are often referred to as "nutri-cereals" because they are packed with vitamins, dietary fibre, and essential minerals like iron and calcium. Because they are native plants, they require less water and can survive in harsh climates, making them great for the environment and farmers.

#### Diagram



#### Real-Life Example

Instead of having regular wheat *roti*, eating a *bajra roti* in winter or a *ragi dosa* for breakfast provides a much higher boost of calcium and iron to keep bones strong.

#### ☒ Quick Revision Checklist:

- ☐ Millets include *jowar*, *bajra*, *ragi*, and *sanwa*.
- ☐ They are extremely rich in minerals and dietary fibre.
- ☐ They are highly adaptable and environmentally friendly crops.

### Topic 7: Food Miles: From Farm to Plate

#### Definition

**Food Miles** refers to the total distance a food item travels from the place it is produced (the farm) to the consumer (your plate).

#### Concept Explanation

Food doesn't just magically appear in the kitchen. For a simple wheat *chapati*, the farmer grows the crop, harvests it, threshes it, stores it, transports it to a mill, grinds it into flour, packs it, transports it to a retail shop, and then we buy it and cook it. The longer this distance (high food miles), the more

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fuel is burnt for transport (causing pollution), and the less fresh the food becomes. Eating locally grown food reduces food miles, helps the environment, supports local farmers, and guarantees fresher meals. Also, considering the immense hard work involved in this chain, we must never waste food!

### Diagram



### Real-Life Example

Buying an apple grown by a farmer in a nearby village has very low food miles compared to buying an apple imported in an airplane all the way from another country.

### ☑ Quick Revision Checklist:

- [ ] Food miles = Distance from farm to plate.
- [ ] Lower food miles mean less pollution and fresher food.
- [ ] We must respect the effort of farmers and avoid wasting food.